L1: Knowledge Check

Each module has an ungraded knowledge check to ensure students comprehend material before moving on to graded assignments and activities.

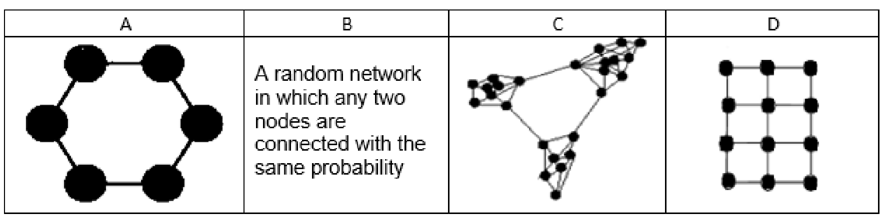
1 Multiple choice 1 point

What is **network science**?

- ◯ None of the above
- ◯ The mathematical study of graph-related data from various disciplines
- ◯ The study of complex systems through their network representation
- ◯ The scientific study of any type of network

2 Multiple choice 1 point

Which one of the following can be described as a **complex** network?



- ◯ D
- ◯ A
- ◯ B
- ◯ C

3 True or False 1 point

Is the following statement **True** or **False**?

The premise of network science is that the network architecture of a complex system is sufficient to understand the system's function and dynamics

- ◯ True
- ◯ False

4 Fill in the Blank 1 point

Which of the following statements is/are **TRUE** regarding network centrality? (Multiple answers)

1. Centrality metrics aim to rank nodes (or edges) based on “importance”. [ ]
2. Although there are various ways (metrics) to define the centrality of nodes and edges, they all give the same node ranking. [ ]
3. In ring networks, all nodes have the same centrality. [ ]
4. Pagerank is a centrality metric but it is only applicable on Web pages. [ ]

- False
- True
- False
- True
- False
- True
- False
- True

5 Multiple choice 1 point

Which of the following statements is **FALSE** regarding communities in networks?

- Communities are clusters of highly interconnected nodes
- The connection density inside a network community is greater than the connection density between network communities
- A network community is a clique (i.e., a fully connected graph)
- Some community detection algorithms assign nodes to more than one community

6 Multiple choice 1 point

In network science, what does the term **“dynamics on networks”** mean?

- The process through which the network topology changes with time
- The process through which the state of network nodes (or links) changes with time, even if the network topology remains the same
- The process through which both the network topology and the state of network nodes (or links) changes with time
- The process through which new edges are added between network nodes

7 Multiple choice 1 point

In Dr. Barabasi’s TED talk, he mentions that scientists expected that the number of FDA-approved drugs would soar after the completion of the Human Genome project. However the reality was the opposite: the number of new FDA-approved drugs per year has decreased since then. How did he explain this effect?

- Because we do not understand the interactions between different genes
- Because the genome is still not mapped very accurately
- None of the above
- Because we do not have a good “part list” for all human genes